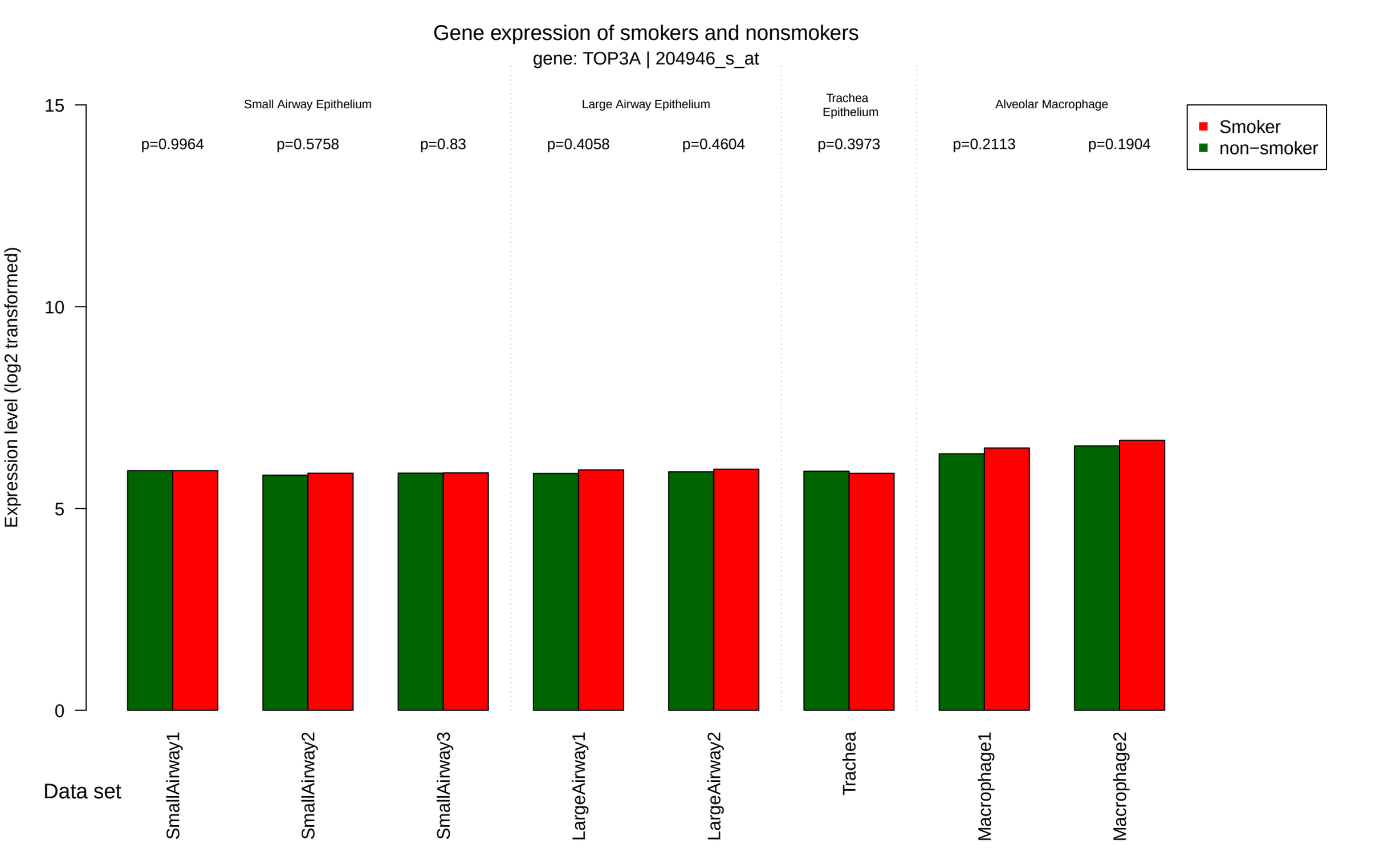
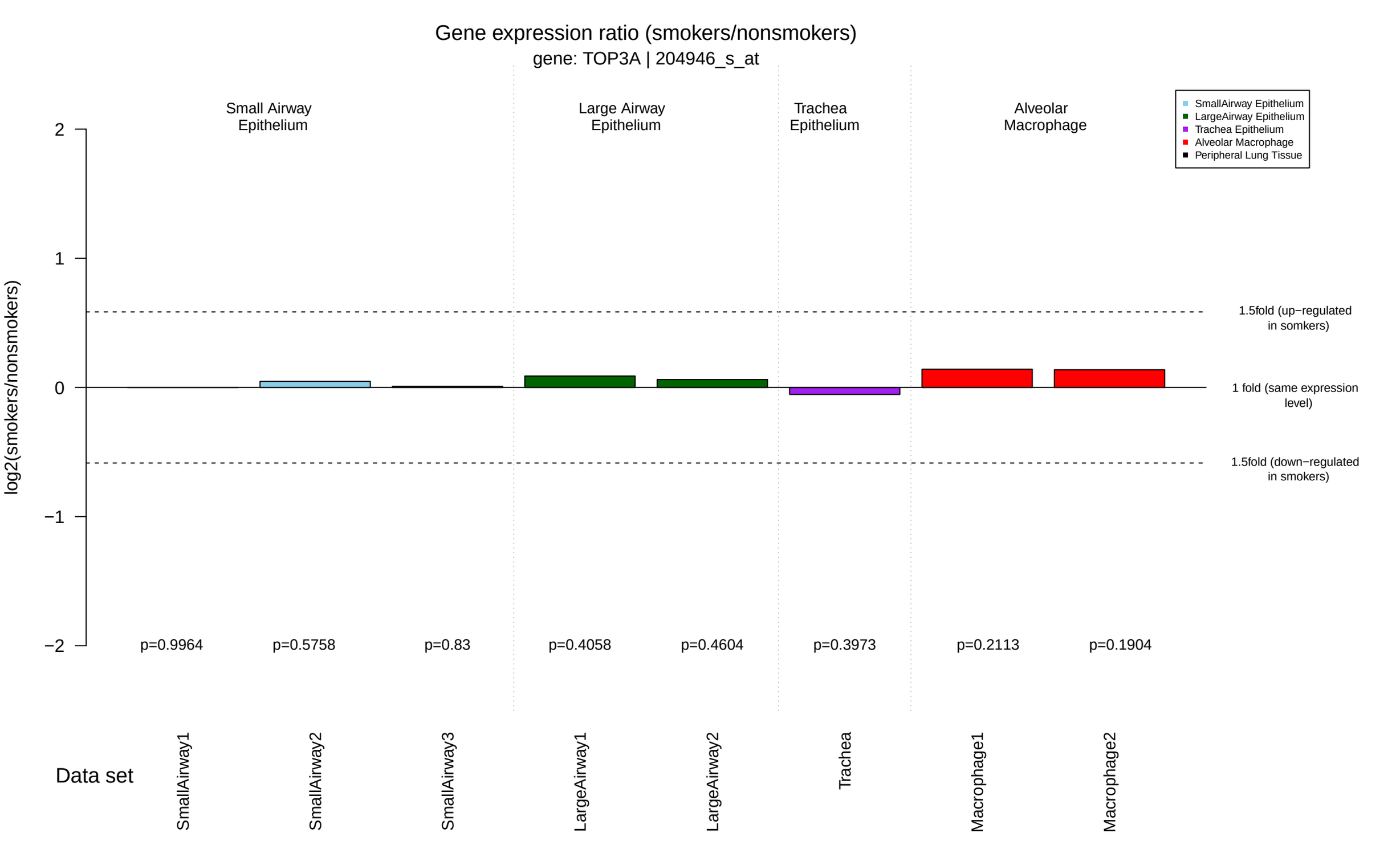
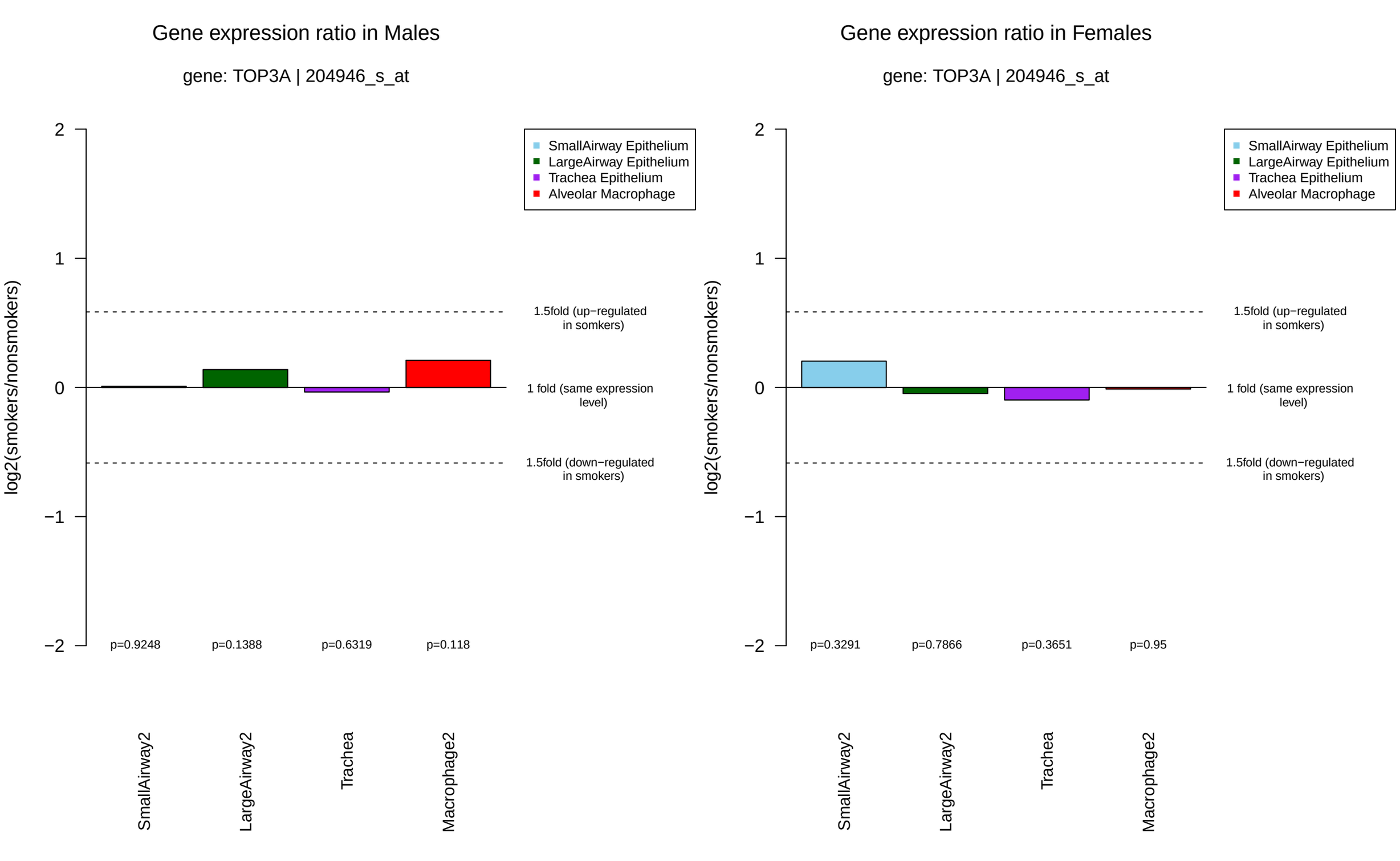
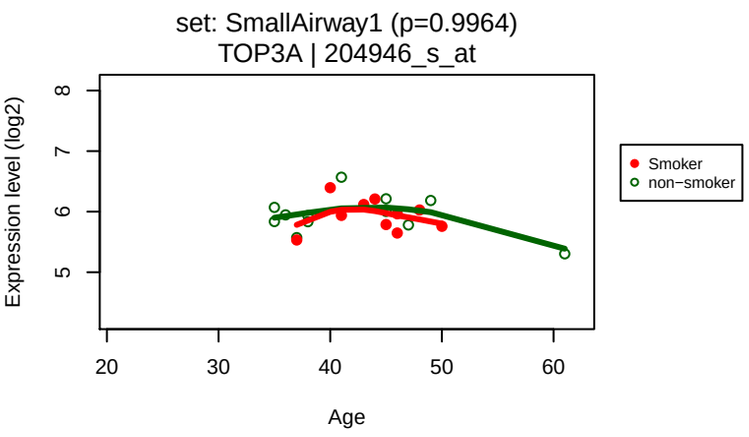
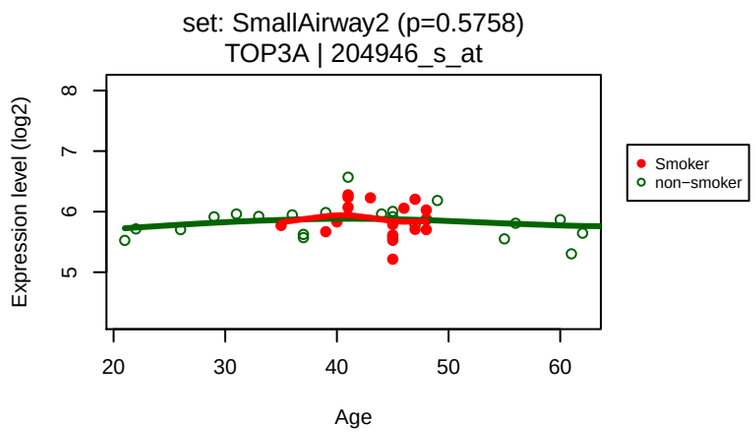
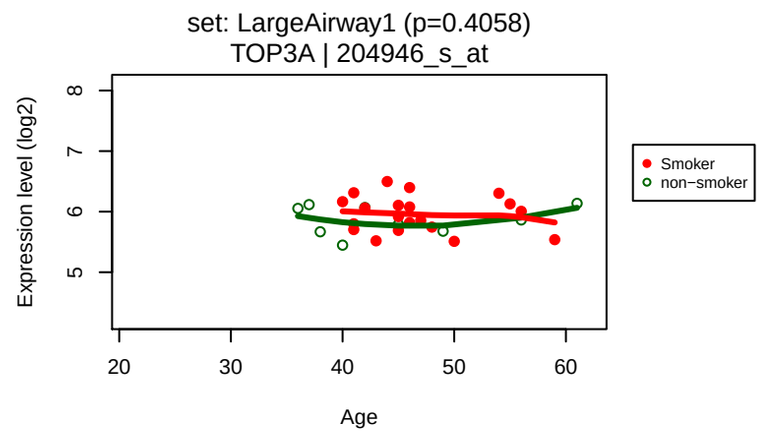

Gene expression of smokers and nonsmokers
gene: TOP3A | 204946_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9964
p=0.5758
p=0.83
p=0.4058
p=0.4604
p=0.3973
p=0.2113
p=0.1904
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

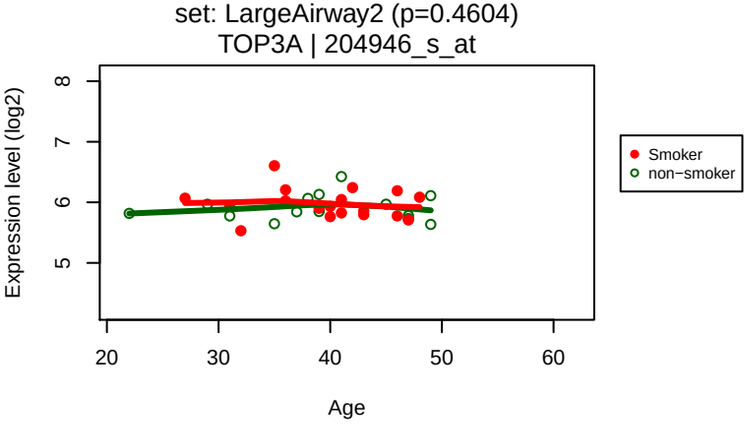

Gene expression ratio (smokers/nonsmokers)
gene: TOP3A | 204946_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9964
p=0.5758
p=0.83
p=0.4058
p=0.4604
p=0.3973
p=0.2113
p=0.1904
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

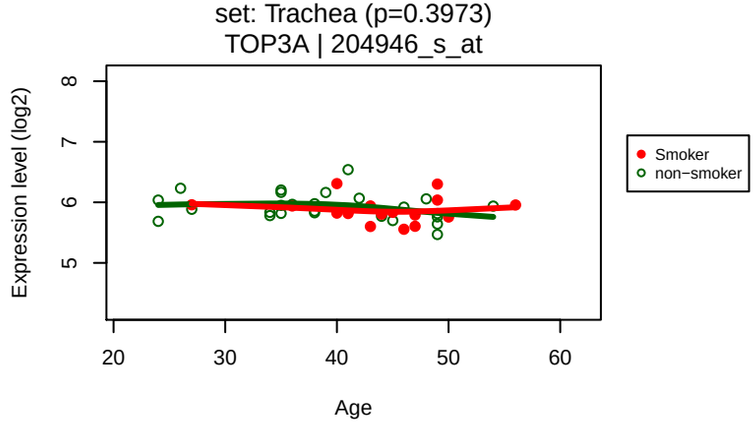

Gene expression ratio in Males
gene: TOP3A | 204946_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9248
p=0.1388
p=0.6319
p=0.118
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TOP3A | 204946_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3291
p=0.7866
p=0.3651
p=0.95
SmallAirway2
LargeAirway2
Trachea
Macrophage2

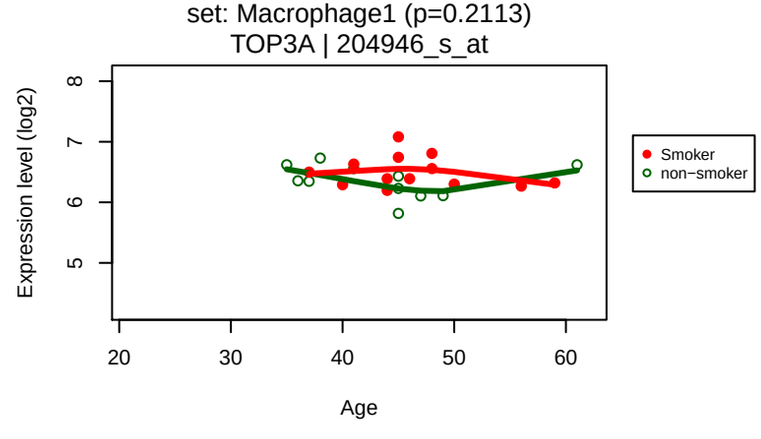

set: SmallAirway1 (p=0.9964)
TOP3A | 204946_s_at
Expression level (log2)
Age
Smoker
non−smoker

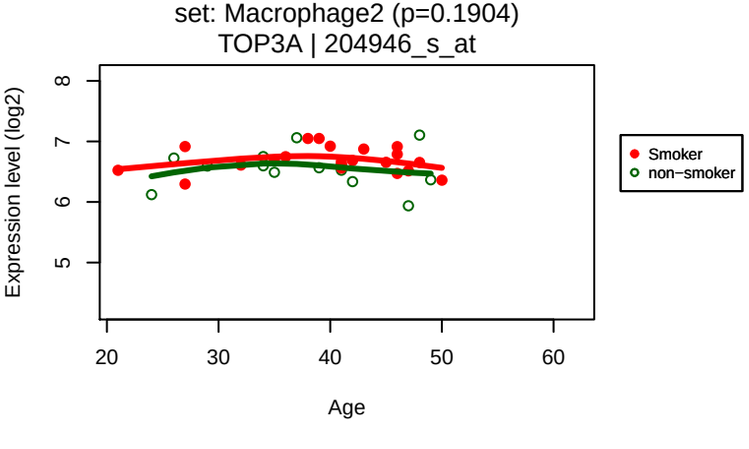

set: SmallAirway2 (p=0.5758)
TOP3A | 204946_s_at
Expression level (log2)
Age
Smoker
non−smoker

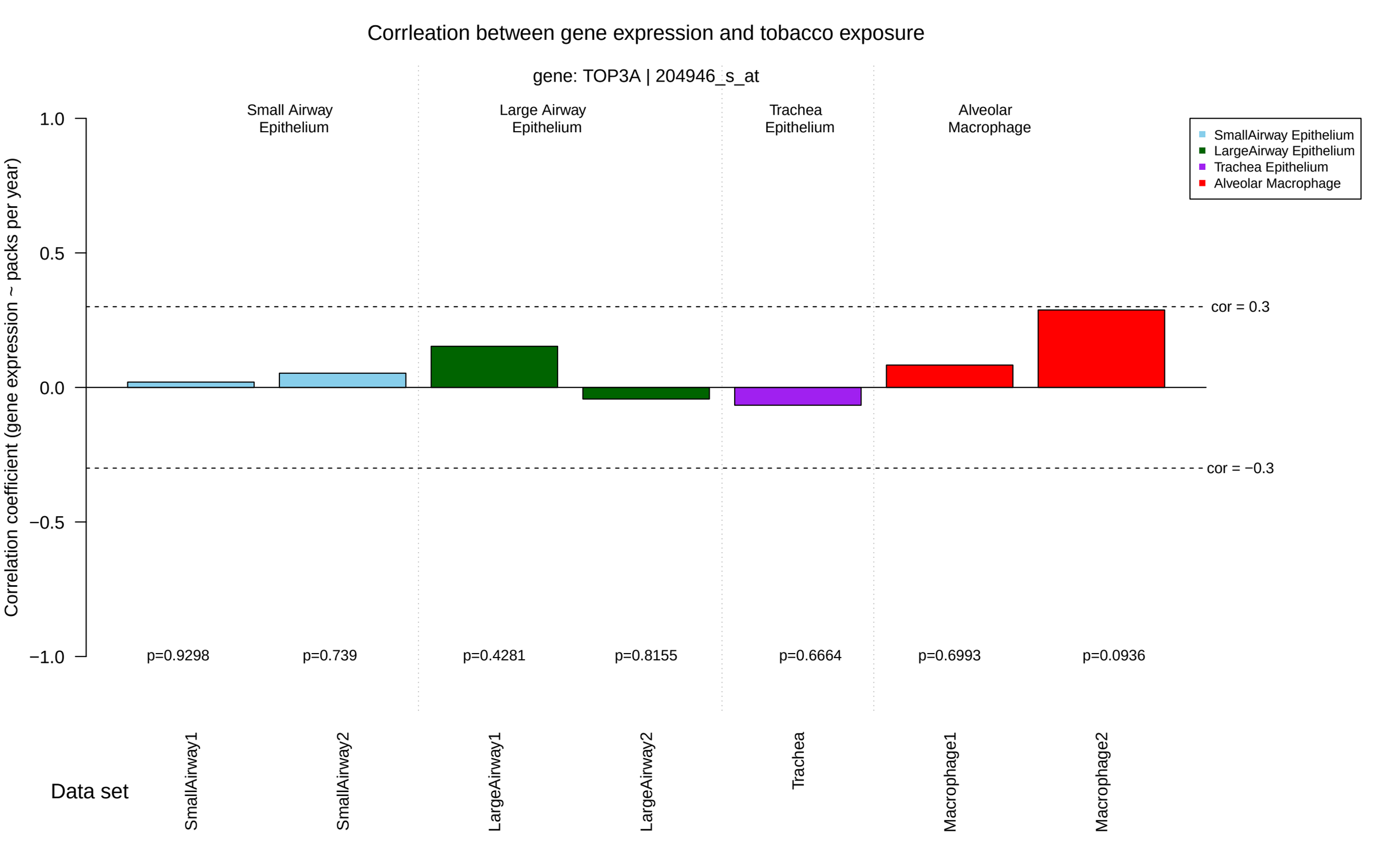

set: LargeAirway1 (p=0.4058)
TOP3A | 204946_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4604)
TOP3A | 204946_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3973)
TOP3A | 204946_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2113)
TOP3A | 204946_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1904)
TOP3A | 204946_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TOP3A | 204946_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9298
p=0.739
p=0.4281
p=0.8155
p=0.6664
p=0.6993
p=0.0936
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2